An Evidence-Centric AI Pipeline for Protein–Disease Regulation Identification

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Introduction

Proteins play fundamental roles in modulating disease initiation, progression, and therapeutic response. The biomedical literature contains extensive experimentally validated information describing these regulatory mechanisms. However, extracting and synthesising such evidence remains a significant challenge due to the growing volume and complexity of scientific publications.

Traditional literature mining approaches primarily detect protein–disease associations without determining regulatory direction or biological context. Furthermore, many AI-driven methods rely on predictive modelling, limiting interpretability and experimental traceability.

Research Gap

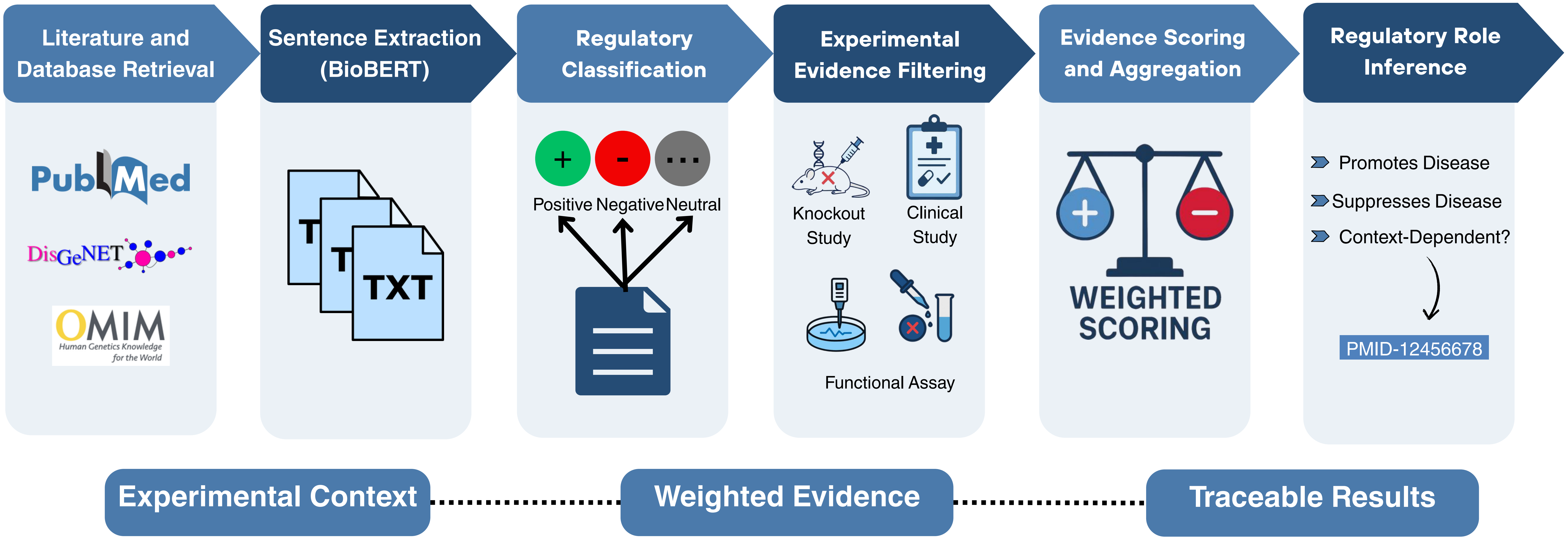
Current literature mining approaches mainly identify protein–disease associations but often fail to determine the direction of regulation and experimental context. Many AI-driven methods rely on predictive or black-box models, limiting interpretability and traceability to validated biological evidence. With the rapid growth of biomedical literature, there is a clear need for transparent, scalable, and evidence-centric frameworks that can systematically extract and synthesize experimentally validated regulatory insights.

Objective

The goal of this study is to develop an evidence-centric AI-enabled framework that:

- Extracts protein–disease regulatory evidence from biomedical literature using domain-specific NLP.
- Classifies regulatory direction based on experimentally supported statements.
- Integrates rule-based experimental filtering to improve biological relevance.
- Aggregates weighted evidence across studies to infer consensus regulatory roles.
- Produces traceable and interpretable outputs linked to original literature sources (based on pipeline workflow + scoring + traceability principles).

Methodological Framework

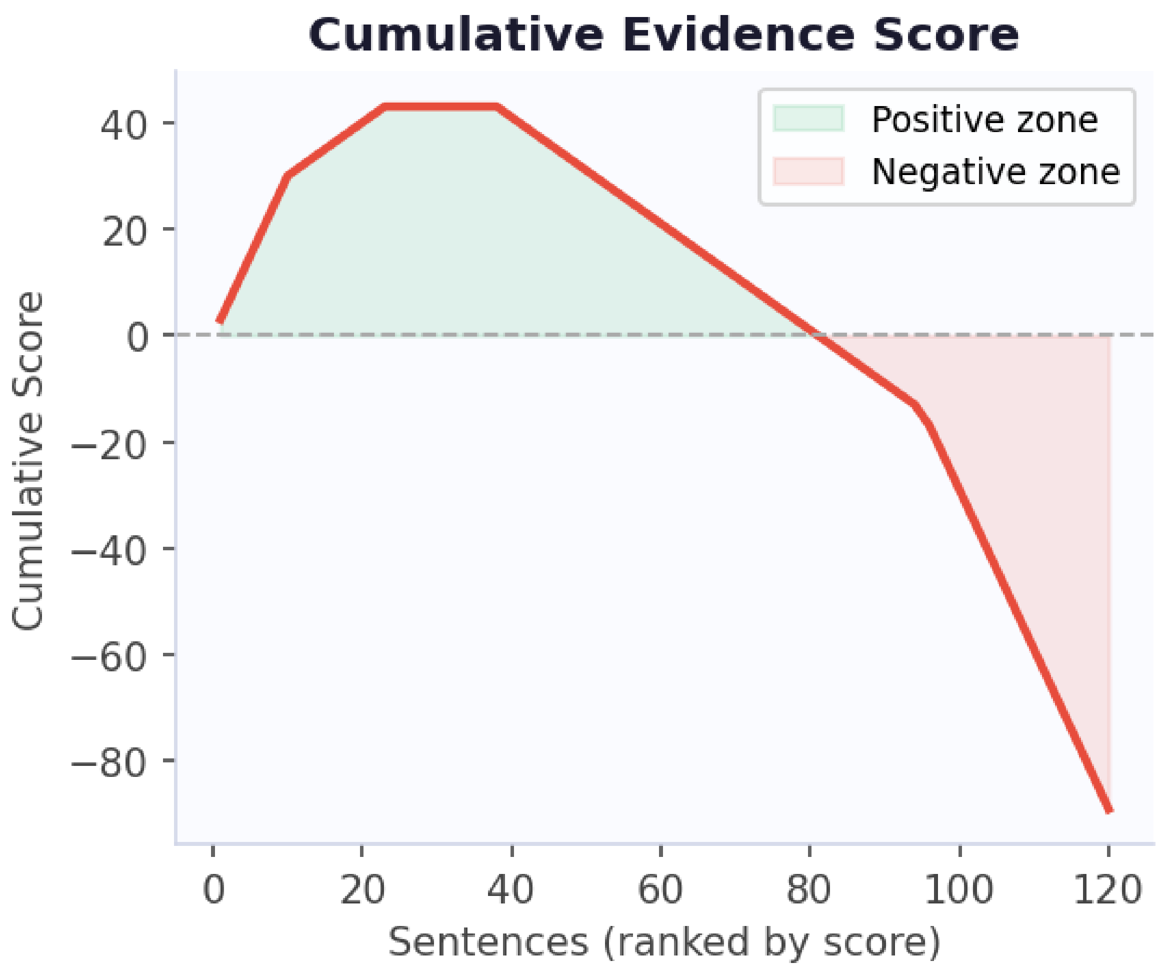


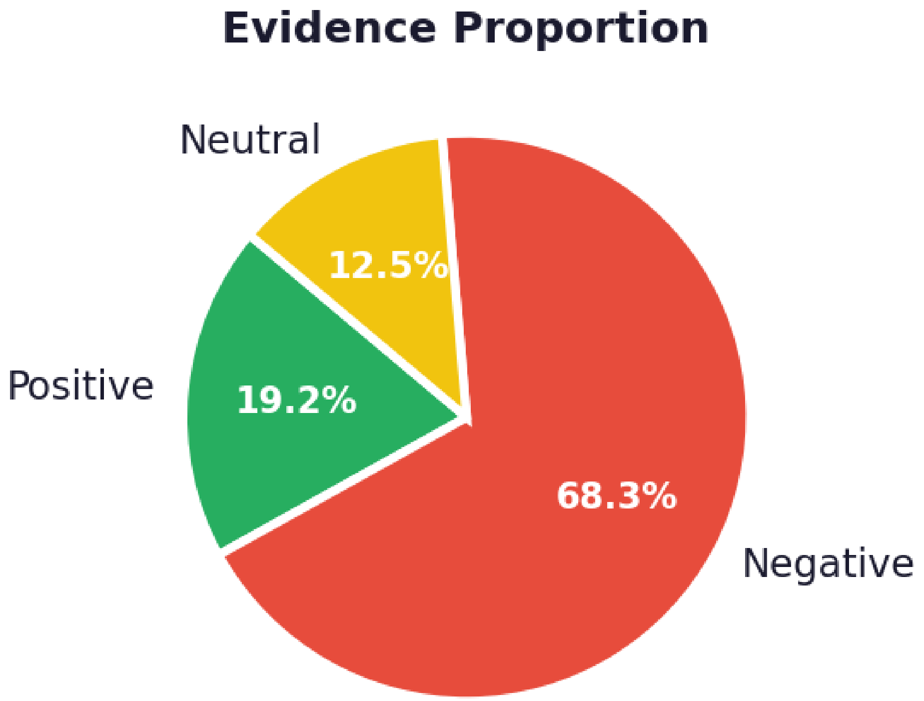
Case Study

Protein: BRCA1

Disease: Breast Cancer

- **Total evidence sentences:** 120
- **Positive:** 23 | **Negative:** 82 | **Neutral:** 15
- **Weighted score**: –89
- **Final role:** Negative regulator (tumour suppressor)





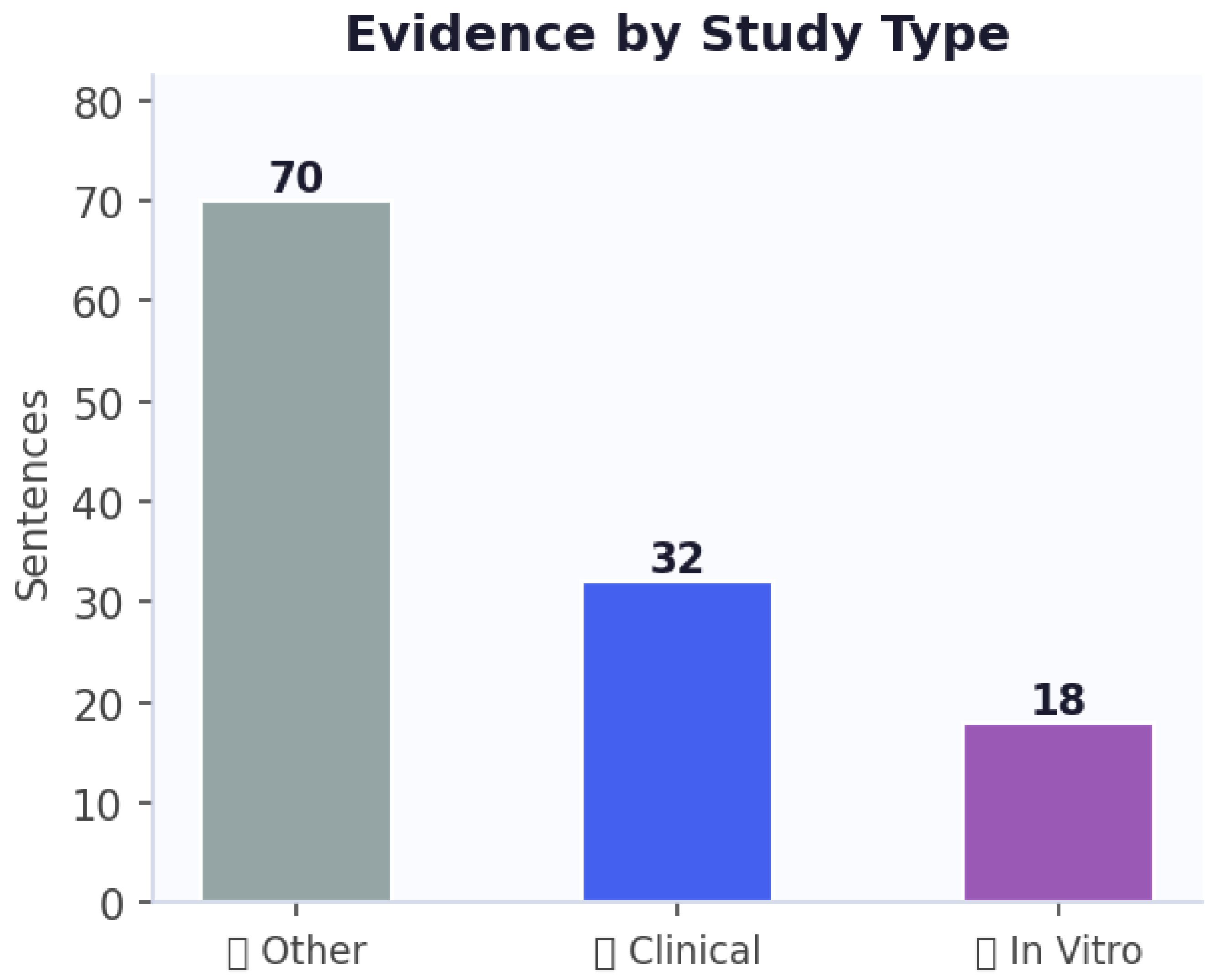
Negative Regulator

The protein SUPPRESSES or protects against the disease based on aggregated weighted evidence from the literature.

Total Score: **-89** | Evidence Sentences: **120** | Positive: **23** | Negative: **82** | Neutral: **15**

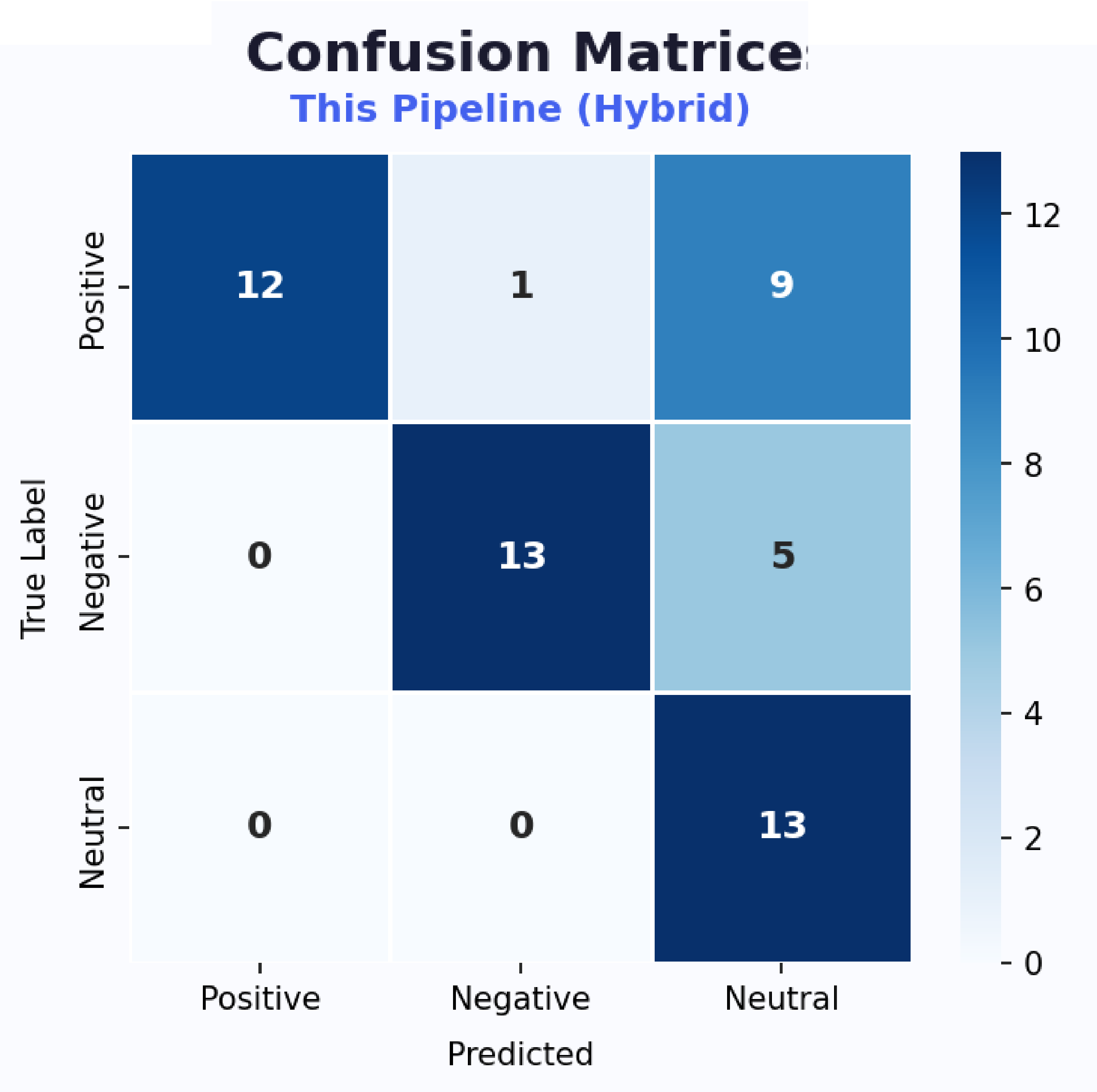
Scalable Evidence Extraction from Biomedical Literature

- **Literature corpus screened:** 248 PubMed abstracts
- **Extracted protein–disease co-mention sentences:** 156
- **Experimentally validated regulatory evidence retained:** 120



Preliminary Direction Classification Performance

F1 score: 0.723
Initial Evaluation on curated regulatory sentence dataset.

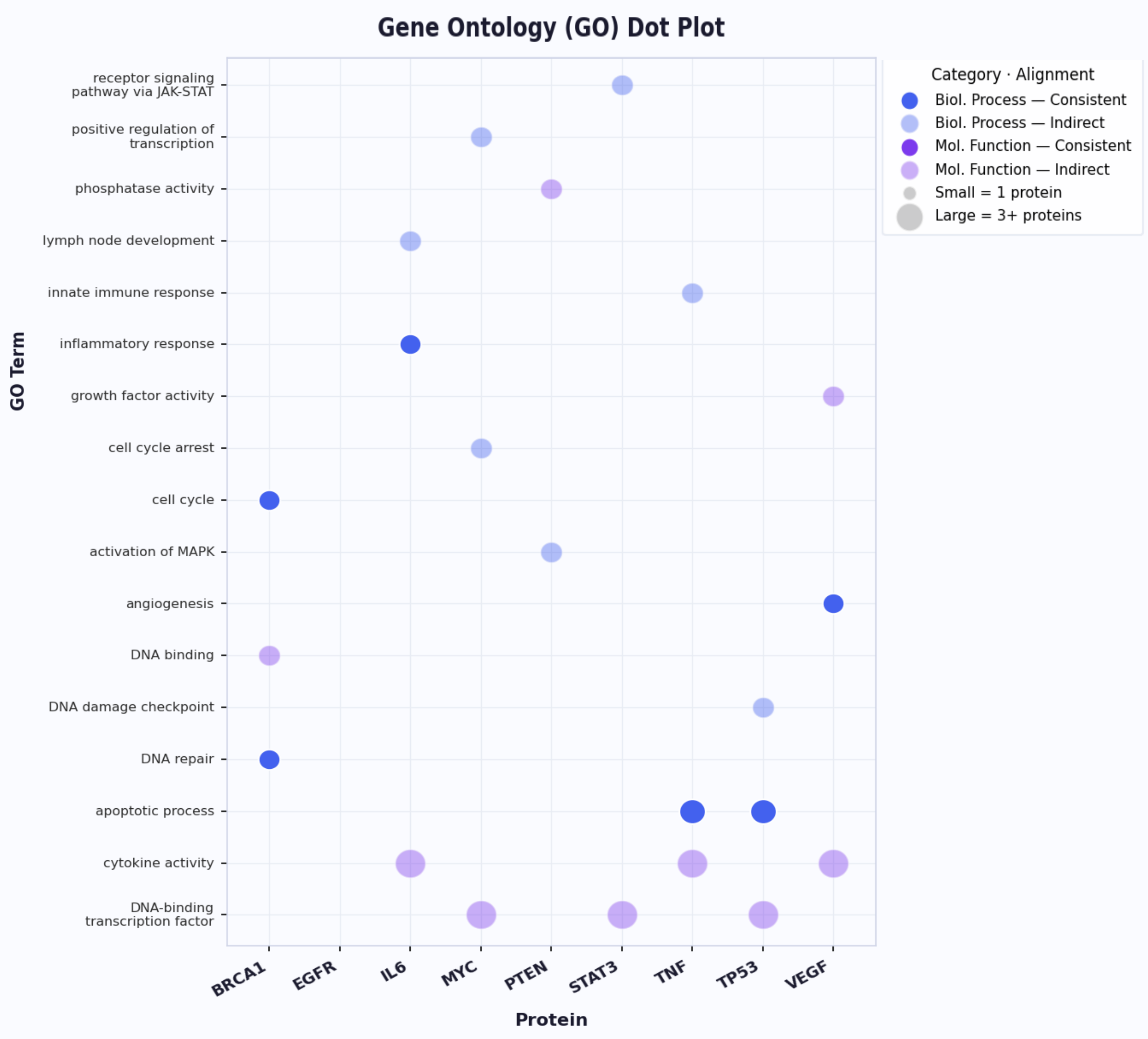


Biological Plausibility of Inferred Regulatory Roles

Predicted regulatory roles demonstrate consistency with known biological mechanisms:

- BRCA1 → DNA repair pathways (tumor suppression)
- IL6 → pro-inflammatory regulation

This supports the framework's ability to capture mechanistically meaningful regulatory relationships.



Conclusion and Future Directions

- Developed an interpretable evidence-centric AI framework
- Enables direction-aware protein–disease inference from literature
- Improves scalability of biological knowledge discovery
- Demonstrates strong computational and biological validation

Future work:

- Integration of full-text biomedical literature mining
- Development of context-aware regulatory reasoning models
- Expansion to multi-omics and pathway-level validation
- Extension towards large-scale regulatory network reconstruction

Thank You